Networks and Causal Inference Short Course: Part 3

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Using Data from the Transmission Reduction Intervention Project in Athens, Greece

Download the data and load it into R Studio

This is a simulated toy dataset made for this course. It is based on data collected from the Transmission Reduction Intervention Project in Athens Greece. Public data here: https://www.icpsr.umich.edu/web/NAHDAP/studies/39059/versions/V2/staff

Read more: https://pubmed.ncbi.nlm.nih.gov/27917890/

Load packages and function script...

```
# -----------------------------
# 1. Read data and functions into R
# -----------------------------

library(lme4)
library(plyr)
library(dplyr)
library(igraph)
library(numDeriv)
library(gtools)
library(doParallel)
library(foreach)
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```
library(kableExtra)
library(ggplot2)
library(forcats)
rm(list = ls())


if ("package:sna" %in% search()) {
  detach("package:sna", unload = TRUE)
}   # sna package is used in part 2, but it will mess up with igraph function in part 3, so I unload it

no_cores <- detectCores() - 1
registerDoParallel(cores=no_cores)
```

```
#source("https://github.com/uri-ncipher/Nearest-Neighbor-estimators/blob/main/functions.R?raw=TRUE")
#source("IPW_functions.R")
#source("gcomp_dr_v2.R") #Ting-Fang removed warnings for bootstrap

# Source required R scripts directly from GitHub
# source("https://raw.githubusercontent.com/uri-ncipher/Causal-Inference-and-Networks-Short-Course/refs/
source("https://raw.githubusercontent.com/uri-ncipher/Causal-Inference-and-Networks-Short-Course/refs/he

### load local IPW_functions.R in order to adding cut to IPW1 propensity score
source("./IPW_functions.R")
```

Data Preparation

Read in the synthetic nodes and edges data sets for creating the simulated network.

```
# ------------------------------
# 2. Data Preparation
# ------------------------------

nodes <- read.csv("https://raw.githubusercontent.com/uri-ncipher/Causal-Inference-and-Networks-Short-Cou
edges <- read.csv("https://raw.githubusercontent.com/uri-ncipher/Causal-Inference-and-Networks-Short-Cou

head(nodes)
```

```
##   id num_nn component posneg date share education employment age avg_posneg
## 1  1      4         1      1    1     1         0          1  39  1.0000000
## 2  2      3         1      1    0     0         1          0  25  0.6666667
## 3  3      1         1      1    0     1         0          1  40  1.0000000
## 4  4      3         1      0    1     1         1          0  37  0.6666667
## 5  5      4         1      1    0     1         1          0  28  0.5000000
## 6  6      2         1      1    1     0         0          1  17  1.0000000
##    avg_date avg_share avg_education avg_employment avg_age     raneff treatment
## 1 0.5000000 0.5000000     0.7500000      0.5000000   27.75 -0.1141261         0
## 2 1.0000000 1.0000000     0.6666667      0.6666667   35.67 -0.1141261         0
## 3 0.0000000 1.0000000     0.0000000      1.0000000   31.00 -0.1141261         1
## 4 0.3333333 0.3333333     0.6666667      0.3333333   29.00 -0.1141261         0
## 5 0.7500000 0.5000000     0.5000000      0.2500000   36.75 -0.1141261         0
## 6 1.0000000 0.5000000     0.5000000      0.5000000   38.00 -0.1141261         1
##   num_tnn num_utnn outcome
## 1       0        4       1
## 2       0        3       0
## 3       0        1       1
## 4       0        3       0
## 5       1        3       1
## 6       1        1       0
```

```
head(edges)
```

```
##   from to
## 1    1  2
## 2    2  4
```

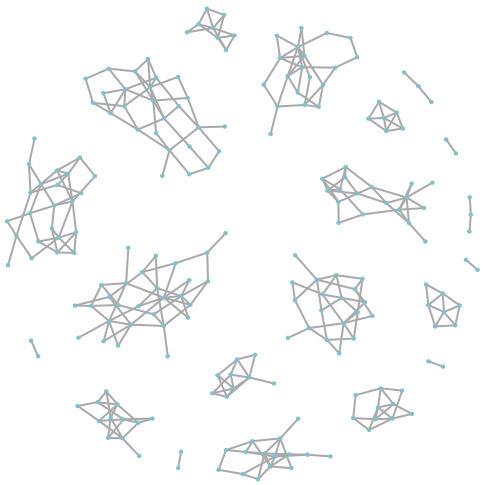
```
## 3    5  8
## 4    1  9
## 5    3  9
## 6    1 10
```

```
net0=graph_from_data_frame(d=edges, vertices = nodes, directed=F)  # create network based on the nodes
```

Network visualization

```
# ------------------------------
# 3. Network Visualizaiton
# ------------------------------

# plot the network generated above
plot(net0, vertex.size=1, vertex.label=NA, vertex.color="cadetblue3", vertex.frame.color="cadetblue3")
```



Prepare the data for modeling

```
# ------------------------------
# 3. Data Preparation for Modeling
# ------------------------------
```

```
# Save the number of individuals (nodes) and number of components
n=length(V(net0))       # number of individuals
m=components(net0)$no   # number of components

# generate a dataset with three columns, the first column is participant id (node id),
# the second column is number of nearest neighbors for each participant (is also node degree),
# the third column indicate which component the participant is in.
df=data.frame(id=1:n, na=unlist(lapply(1:n, num_neighbors, net=net0)),
              component=components(net0)$membership)

# assign treatment, outcome, and baseline covariates to data
df$treatment=nodes$treatment
df$outcome=nodes$outcome
df$age=nodes$age
df$age10=nodes$age/10 #age per 10 years
df$agebin=ifelse(nodes$age>33,1,0) #age >33 years old
table(df$treatment, df$agebin)
```

```
##
##       0   1
##   0  82 106
##   1  15  13
```

```
df$posneg=nodes$posneg
df$date=nodes$date
df$share=nodes$share
df$education=nodes$education
df$employment=nodes$employment
df$na_a= unlist(lapply(1:n, trt_neighbors, net=net0))
df$notna_a=df$na-df$na_a

# six covariates, including age
base_covariate=c("age10", "posneg", "date", "share", "education", "employment")

# averaging baseline covariates (average of nearest neighbors' baseline covariate values)
# Note: show code for averaging "age10" and "agebin" as examples. Average of other neighbor covariates
#       Example code: df$avg_var1=unlist(lapply(1:n, avg_neighbors, net=net0, variable="var1")) # "var1
df$avg_agebin=unlist(lapply(1:n, avg_neighbors, net=net0, variable="agebin"))
df$avg_age10 =unlist(lapply(1:n, avg_neighbors, net=net0, variable="age10"))

df$avg_posneg = nodes$avg_posneg
df$avg_date = nodes$avg_date
df$avg_share = nodes$avg_share
df$avg_education = nodes$avg_education
df$avg_employment = nodes$avg_employment

avg_covariate=c("avg_age10","avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")
```

Modeling

In this section, we will apply both IPW1 and IPW2 estimators for the average potential outcomes and corresponding causal effects. An individual's exposure/treatment and allocation strategy α jointly determine the

average potential outcome (allocation strategy represents the probability that individuals in nearest neighbors set receiving the exposure/treatment in the counterfactual scenario, so it is a numeric value between 0 and 1). For each estimator, the R programs will output the **point estimation and the estimated variance of average potential outcomes $\widehat{\boldsymbol{Y}}(\mathbf{1},\boldsymbol{\alpha}), \widehat{\boldsymbol{Y}}(\mathbf{0},\boldsymbol{\alpha}), \widehat{\boldsymbol{Y}}(\boldsymbol{\alpha})$** under three different allocation strategies α (i.e., 0.25, 0.50, and 0.75), respectively. Also, the R program will output the **point estimation and estimated variance of four causal effects: Direct** (DE), **Indirect** (IE), **Total** (TE), and **Overall effect** (OE).

Equations for calculating the four causal effects are:

- $\widehat{DE} = \widehat{Y}(1,\alpha) - \widehat{Y}(0,\alpha)$
- $\widehat{IE} = \widehat{Y}(0,\alpha_0) - \widehat{Y}(0,\alpha_1)$
- $\widehat{TE} = \widehat{Y}(1,\alpha_0) - \widehat{Y}(0,\alpha_1)$
- $\widehat{OE} = \widehat{Y}(\alpha_0) - \widehat{Y}(\alpha_1)$,

where α_0, α_1 both represents allocation strategies, and $\alpha_0 \neq \alpha_1$ (Note: The associated paper compared the estimated average potential outcomes with α_1 to α_0 for the indirect, total, and overall effects. The code here compared the estimated average potential outcomes with α_0 to α_1 for the indirect, total, and overall effects). Users can set any values between 0 and 1 for α_0 and α_1. If given α is a vector, then the causal contrasts will produce pairwise comparisons in the sequential order of the values in α. For example, if $\alpha = c(0.75, 0.25, 0.50)$, the contrasts for spillover effects will compare

- (i) $\alpha_0 = 0.75$ versus $\alpha_1 = 0.25$;
- (ii) $\alpha_0 = 0.75$ versus $\alpha_1 = 0.50$;
- (iii) $\alpha_0 = 0.25$ versus $\alpha_1 = 0.50$.

IPW1

Using IPW1 to estimate the average potential outcomes and causal effects under allocation strategies α.

```
# ------------------------------
# 4a. IPW1 Estimator
# ------------------------------

alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

base_covariate=c("agebin", "posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_agebin","avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")
#Note: Age variable continuous removed due to convergence issue, used binary for now

IPW1 <- IPW_1_model(df, base_covariate, alpha)
```

```
## [1] "summary(score):"
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## 0.0004355 0.0857978 0.5219730 0.4342091 0.6918815 0.8869194
```

```
IPW1
```

```
## [[1]]
##          a=1         a=0      margin alpha           type
## 1 0.25529569 0.360633818 0.307964755   0.5 point estimate
```

```
## 2 0.35299392 0.409391168 0.386832268   0.4 point estimate
## 3 0.47297810 0.451569373 0.457991992   0.3 point estimate
## 4 0.60178904 0.480892919 0.505072143   0.2 point estimate
## 5 0.01197303 0.006377590 0.006241201   0.5       variance
## 6 0.02225333 0.006663655 0.008690831   0.4       variance
## 7 0.03733800 0.007352668 0.010751031   0.3       variance
## 8 0.05183687 0.008203765 0.011392133   0.2       variance
##
## [[2]]
##       estimation alpha0 alpha1     type
## 1  -0.1053381259    0.5    0.5   Direct
## 2  -0.0563972494    0.4    0.4   Direct
## 3   0.0214087314    0.3    0.3   Direct
## 4   0.1208961205    0.2    0.2   Direct
## 5   0.0117364409    0.5    0.5   Var DE
## 6   0.0175362249    0.4    0.4   Var DE
## 7   0.0266535111    0.3    0.3   Var DE
## 8   0.0346140798    0.2    0.2   Var DE
## 9  -0.0487573503    0.5    0.4 Indirect
## 10 -0.0909355550    0.5    0.3 Indirect
## 11 -0.1202591017    0.5    0.2 Indirect
## 12 -0.0421782047    0.4    0.3 Indirect
## 13 -0.0715017514    0.4    0.2 Indirect
## 14 -0.0293235467    0.3    0.2 Indirect
## 15  0.0004454532    0.5    0.4   Var IE
## 16  0.0017234579    0.5    0.3   Var IE
## 17  0.0035909238    0.5    0.2   Var IE
## 18  0.0004310698    0.4    0.3   Var IE
## 19  0.0015848385    0.4    0.2   Var IE
## 20  0.0003772808    0.3    0.2   Var IE
## 21 -0.1540954762    0.5    0.4    Total
## 22 -0.1962736808    0.5    0.3    Total
## 23 -0.2255972276    0.5    0.2    Total
## 24 -0.0985754541    0.4    0.3    Total
## 25 -0.1278990008    0.4    0.2    Total
## 26 -0.0079148153    0.3    0.2    Total
## 27  0.0097343132    0.5    0.4   Var TE
## 28  0.0084611152    0.5    0.3   Var TE
## 29  0.0080638593    0.5    0.2   Var TE
## 30  0.0155535556    0.4    0.3   Var TE
## 31  0.0146177626    0.4    0.2   Var TE
## 32  0.0251110409    0.3    0.2   Var TE
## 33 -0.0788675135    0.5    0.4  Overall
## 34 -0.1500272373    0.5    0.3  Overall
## 35 -0.1971073887    0.5    0.2  Overall
## 36 -0.0711597239    0.4    0.3  Overall
## 37 -0.1182398753    0.4    0.2  Overall
## 38 -0.0470801514    0.3    0.2  Overall
## 39  0.0003716931    0.5    0.4   Var OE
## 40  0.0012527940    0.5    0.3   Var OE
## 41  0.0023005983    0.5    0.2   Var OE
## 42  0.0002950416    0.4    0.3   Var OE
## 43  0.0010944169    0.4    0.2   Var OE
## 44  0.0003338455    0.3    0.2   Var OE
```

In the output above,

- The **section** [[**1**]] displays the point estimates (type = "point estimate" in the output) and estimated variances (type = "variance" in the output) for the average potential outcomes under three different allocation strategies (i.e., $\alpha = 0.25, 0.50, 0.75$) and each individual exposure ($a = 0, a = 1$, and margin). "Margin" is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.
- The **section** [[**2**]] displays the point estimates (type = "Direct", "Indirect", "Total", "Overall" in the output) and estimated variances (type = "Var DE", "Var IE", "Var TE", "Var OE" in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies (α_0 and α_1). The estimated values are shown in "estimation" column in the output above.

Goodness of Fit test for random effect in mixed effect model.

```
### install glmerGOF package (github link: https://github.com/BarkleyBG/glmerGOF)
# remotes::install_github("BarkleyBG/glmerGOF")

library(glmerGOF)
library(survival)
library(cubature) # testGOF() function requires this package
```

```
## Warning: package 'cubature' was built under R version 4.4.3
```

```
### Fit a lme4::glmer() model (mixed effect model)
formula_glmer <- as.formula(paste("treatment ~", paste(base_covariate, collapse = "+"), "+ (1 | componer
formula_glmer
```

```
## treatment ~ agebin + posneg + date + share + education + employment +
##     (1 | component)
```

```
fit_glmer <- glmer(formula_glmer, data=df, family=binomial(link="logit"))


### Fit a survival::clogit() model
formula_clogit <- as.formula(paste("treatment ~", paste(base_covariate, collapse = "+"), "+ strata(compo
formula_clogit
```

```
## treatment ~ agebin + posneg + date + share + education + employment +
##     strata(component)
```

```
fit_clogit <- survival::clogit(formula_clogit, data=df, method="exact")

variable_names <- list(DV="treatment", grouping="component")

TC_test <- testGOF(data=df,
                   fitted_model_clogit = fit_clogit,
                   fitted_model_glmm = fit_glmer,
                   var_names = variable_names,
                   gradient_derivative_method = "simple"
                   )
TC_test$results
```

```
## $D
## [1] -5.604119
##
## $p_value
## [1] 1
##
## $df
## [1] 6
```

```
# ### Shapiro-Wilk normality test
# re <- ranef(fit_glmer)
# re_values <- re$component[["(Intercept)"]]
# shapiro.test(re_values)
#
# ### Lilliefors (Kolmogorov-Smirnov) test for normality
# library(nortest)
# nortest::lillie.test(re_values)
```

IPW2

Using IPW2 to estimate the average potential outcomes and causal effects under allocation strategies α.

```
# ------------------------------
# 4b. IPW2 Estimator
# ------------------------------
#change back to age per 10 year increase
base_covariate=c("age10", "posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_age10","avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")

alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

formula_1=paste("cbind(na_a, notna_a)", "~", "treatment", "+",
                paste(base_covariate, collapse = "+"), "+",
                paste(avg_covariate, collapse = "+"))
formula_2=paste("treatment", "~", paste(base_covariate, collapse = "+"))
M1=glm(formula_1, family = binomial(link = "logit"), data=df)
M2=glm(formula_2, family = binomial(link = "logit"), data=df)

IPW2 <- IPW_2_model(df, M1, M2, alpha)
IPW2
```

```
## [[1]]
##           a=1        a=0     margin alpha           type
## 1 0.220810727 0.59176096 0.40628585   0.5 point estimate
## 2 0.292392238 0.59042302 0.47121071   0.4 point estimate
## 3 0.387385956 0.57233698 0.51685167   0.3 point estimate
## 4 0.505408711 0.54206916 0.53473707   0.2 point estimate
## 5 0.006943618 0.09001969 0.03029339   0.5       variance
## 6 0.012269497 0.05617945 0.02906814   0.4       variance
## 7 0.023480049 0.02942577 0.02347725   0.3       variance
## 8 0.044905869 0.01507811 0.01736360   0.2       variance
##
## [[2]]
```

```
##       estimation alpha0 alpha1     type
## 1  -0.3709502369    0.5    0.5   Direct
## 2  -0.2980307830    0.4    0.4   Direct
## 3  -0.1849510226    0.3    0.3   Direct
## 4  -0.0366604489    0.2    0.2   Direct
## 5   0.0727530662    0.5    0.5   Var DE
## 6   0.0397805488    0.4    0.4   Var DE
## 7   0.0198324068    0.3    0.3   Var DE
## 8   0.0230003806    0.2    0.2   Var DE
## 9   0.0013379431    0.5    0.4 Indirect
## 10  0.0194239855    0.5    0.3 Indirect
## 11  0.0496918036    0.5    0.2 Indirect
## 12  0.0180860424    0.4    0.3 Indirect
## 13  0.0483538604    0.4    0.2 Indirect
## 14  0.0302678181    0.3    0.2 Indirect
## 15  0.0050022015    0.5    0.4   Var IE
## 16  0.0222549084    0.5    0.3   Var IE
## 17  0.0485650418    0.5    0.2   Var IE
## 18  0.0061643281    0.4    0.3   Var IE
## 19  0.0224483697    0.4    0.2   Var IE
## 20  0.0050978758    0.3    0.2   Var IE
## 21 -0.3696122938    0.5    0.4    Total
## 22 -0.3515262514    0.5    0.3    Total
## 23 -0.3212584334    0.5    0.2    Total
## 24 -0.2799447407    0.4    0.3    Total
## 25 -0.2496769226    0.4    0.2    Total
## 26 -0.1546832045    0.3    0.2    Total
## 27  0.0412503517    0.5    0.4   Var TE
## 28  0.0175815581    0.5    0.3   Var TE
## 29  0.0064830895    0.5    0.2   Var TE
## 30  0.0170154604    0.4    0.3   Var TE
## 31  0.0069144403    0.4    0.2   Var TE
## 32  0.0110807476    0.3    0.2   Var TE
## 33 -0.0649248621    0.5    0.4  Overall
## 34 -0.1105658262    0.5    0.3  Overall
## 35 -0.1284512251    0.5    0.2  Overall
## 36 -0.0456409641    0.4    0.3  Overall
## 37 -0.0635263630    0.4    0.2  Overall
## 38 -0.0178853989    0.3    0.2  Overall
## 39  0.0004946305    0.5    0.4   Var OE
## 40  0.0032369653    0.5    0.3   Var OE
## 41  0.0100291441    0.5    0.2   Var OE
## 42  0.0013422311    0.4    0.3   Var OE
## 43  0.0067075875    0.4    0.2   Var OE
## 44  0.0021150165    0.3    0.2   Var OE
```

In the output above,

- The **section** **[[1]]** displays the point estimates (type = "point estimate" in the output) and estimated variances (type = "variance" in the output) for the average potential outcomes under three different allocation strategies (i.e., $\alpha = 0.75, 0.50, 0.25$) and each individual exposure ($a = 0, a = 1$, and margin). "Margin" is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.

- The **section** [[**2**]] displays the point estimates (type = "Direct", "Indirect", "Total", "Overall" in the output) and estimated variances (type = "Var DE", "Var IE", "Var TE", "Var OE" in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies (α_0 and α_1). The estimated values are shown in "estimation" column in the output above.

Now, we provide example code for an outcome-model based approach, g-computation. We use assumptions similar to IPW2 regarding the covariates for confounding from an individual's contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts' exposures and covariates.

```
# ------------------------------------------------------------
# 4c. G-computation Estimator (similar assumptions to IPW2)
# ------------------------------------------------------------

#Obtain G-computation estimates with bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts

alpha <- c(0.5, 0.4, 0.3, 0.2)

res_gcomp <- gcomp_with_bootstrap(
  data = df,
  alpha = alpha,
  z_covariates = base_covariate,
  zn_covariates = avg_covariate,
  B = 1000,
  seed = 2026
)

res_gcomp$results
```

```
##       estimation alpha0 alpha1     type       lower      upper
## 1  -0.2942459106    0.5    0.5   Direct -0.50761492 0.06533357
## 2  -0.2293979865    0.4    0.4   Direct -0.42669257 0.06419693
## 3  -0.1598010505    0.3    0.3   Direct -0.33798759 0.07663089
## 4  -0.0854203130    0.2    0.2   Direct -0.25944479 0.12368184
## 5   0.0216511692    0.5    0.5   Var DE          NA         NA
## 6   0.0164000766    0.4    0.4   Var DE          NA         NA
## 7   0.0120227666    0.3    0.3   Var DE          NA         NA
## 8   0.0091832561    0.2    0.2   Var DE          NA         NA
## 9   0.0200839175    0.5    0.4 Indirect -0.01693391 0.04428758
## 10  0.0403768693    0.5    0.3 Indirect -0.03420301 0.09193725
## 11  0.0608319406    0.5    0.2 Indirect -0.05179040 0.14283959
## 12  0.0202929518    0.4    0.3 Indirect -0.01726860 0.04778418
## 13  0.0407480231    0.4    0.2 Indirect -0.03485546 0.09824659
## 14  0.0204550713    0.3    0.2 Indirect -0.01758550 0.05084486
## 15  0.0002468110    0.5    0.4   Var IE          NA         NA
## 16  0.0010415511    0.5    0.3   Var IE          NA         NA
## 17  0.0024635162    0.5    0.2   Var IE          NA         NA
## 18  0.0002745770    0.4    0.3   Var IE          NA         NA
## 19  0.0011522778    0.4    0.2   Var IE          NA         NA
## 20  0.0003021185    0.3    0.2   Var IE          NA         NA
## 21 -0.2741619931    0.5    0.4    Total -0.47240345 0.07597257
```

```
## 22 -0.2538690413    0.5    0.3    Total -0.43353718 0.09322211
## 23 -0.2334139700    0.5    0.2    Total -0.40302065 0.10170925
## 24 -0.2091050347    0.4    0.3    Total -0.39125631 0.07023090
## 25 -0.1886499634    0.4    0.2    Total -0.36128077 0.09155945
## 26 -0.1393459792    0.3    0.2    Total -0.30901130 0.08943311
## 27  0.0196262256    0.5    0.4   Var TE          NA         NA
## 28  0.0180032470    0.5    0.3   Var TE          NA         NA
## 29  0.0168711890    0.5    0.2   Var TE          NA         NA
## 30  0.0148275764    0.4    0.3   Var TE          NA         NA
## 31  0.0137482572    0.4    0.2   Var TE          NA         NA
## 32  0.0110057702    0.3    0.2   Var TE          NA         NA
## 33 -0.0352798432    0.5    0.4  Overall -0.05516482 0.02138685
## 34 -0.0588057709    0.5    0.3  Overall -0.10258781 0.04539049
## 35 -0.0692069521    0.5    0.2  Overall -0.13731253 0.07315086
## 36 -0.0235259276    0.4    0.3  Overall -0.04687556 0.02419022
## 37 -0.0339271089    0.4    0.2  Overall -0.08552364 0.05206213
## 38 -0.0104011812    0.3    0.2  Overall -0.03868068 0.02949115
## 39  0.0003810760    0.5    0.4   Var OE          NA         NA
## 40  0.0013757670    0.5    0.3   Var OE          NA         NA
## 41  0.0027839896    0.5    0.2   Var OE          NA         NA
## 42  0.0003201982    0.4    0.3   Var OE          NA         NA
## 43  0.0011822972    0.4    0.2   Var OE          NA         NA
## 44  0.0002889667    0.3    0.2   Var OE          NA         NA
```

Now, we provide example code for an doubly-robust approach, augmented IPW. We use assumptions similar to IPW2 regarding the covariates for confounding from an individual's contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts' exposures and covariates.

```
# ------------------------------------------------------------
# 4d. Augmented IPW Estimator (doubly robust (DR1))
# ------------------------------------------------------------

#Obtain DR1 Estimates with Bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts
alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

res_dr <- dr_with_bootstrap(
  data = df,
  alpha = alpha,
  z_covariates = base_covariate,
  zn_covariates = avg_covariate,
  B = 1000,
  seed = 2026
)
res_dr$results
```

```
##       estimation alpha0 alpha1     type       lower        upper
## 1  -0.4455634514    0.5    0.5   Direct -0.93715545 -0.016765493
## 2  -0.3577497959    0.4    0.4   Direct -0.76750642  0.030082246
## 3  -0.2539181376    0.3    0.3   Direct -0.56140257  0.094975729
## 4  -0.1342450916    0.2    0.2   Direct -0.39851105  0.232588637
```

```
## 5   0.0531417912    0.5    0.5   Var DE          NA           NA
## 6   0.0414580541    0.4    0.4   Var DE          NA           NA
## 7   0.0390018854    0.3    0.3   Var DE          NA           NA
## 8   0.0526416964    0.2    0.2   Var DE          NA           NA
## 9   0.0449502665    0.5    0.4 Indirect -0.01084133  0.149703682
## 10  0.0918239271    0.5    0.3 Indirect -0.02200590  0.315064980
## 11  0.1374273586    0.5    0.2 Indirect -0.03276185  0.478209199
## 12  0.0468736607    0.4    0.3 Indirect -0.01165771  0.167767222
## 13  0.0924770921    0.4    0.2 Indirect -0.02306165  0.330362955
## 14  0.0456034314    0.3    0.2 Indirect -0.01098028  0.160935258
## 15  0.0016549408    0.5    0.4   Var IE          NA           NA
## 16  0.0074018887    0.5    0.3   Var IE          NA           NA
## 17  0.0165648912    0.5    0.2   Var IE          NA           NA
## 18  0.0020647643    0.4    0.3   Var IE          NA           NA
## 19  0.0077881332    0.4    0.2   Var IE          NA           NA
## 20  0.0018424493    0.3    0.2   Var IE          NA           NA
## 21 -0.4006131849    0.5    0.4    Total -0.80526016 -0.008135616
## 22 -0.3537395242    0.5    0.3    Total -0.66760483  0.031608331
## 23 -0.3081360928    0.5    0.2    Total -0.55378513  0.071303381
## 24 -0.3108761352    0.4    0.3    Total -0.61707076  0.047007159
## 25 -0.2652727037    0.4    0.2    Total -0.49385115  0.083106573
## 26 -0.2083147061    0.3    0.2    Total -0.44164396  0.123325407
## 27  0.0404582776    0.5    0.4   Var TE          NA           NA
## 28  0.0301274437    0.5    0.3   Var TE          NA           NA
## 29  0.0240662094    0.5    0.2   Var TE          NA           NA
## 30  0.0309960585    0.4    0.3   Var TE          NA           NA
## 31  0.0248090126    0.4    0.2   Var TE          NA           NA
## 32  0.0326468813    0.3    0.2   Var TE          NA           NA
## 33 -0.0347315409    0.5    0.4  Overall -0.06821944  0.023101619
## 34 -0.0547823573    0.5    0.3  Overall -0.12702281  0.066721805
## 35 -0.0585053488    0.5    0.2  Overall -0.16705565  0.146756561
## 36 -0.0200508164    0.4    0.3  Overall -0.05813752  0.051277125
## 37 -0.0237738079    0.4    0.2  Overall -0.09998771  0.114006996
## 38 -0.0037229915    0.3    0.2  Overall -0.04554477  0.070645016
## 39  0.0006677682    0.5    0.4   Var OE          NA           NA
## 40  0.0026659983    0.5    0.3   Var OE          NA           NA
## 41  0.0059769886    0.5    0.2   Var OE          NA           NA
## 42  0.0007509683    0.4    0.3   Var OE          NA           NA
## 43  0.0030976513    0.4    0.2   Var OE          NA           NA
## 44  0.0008663912    0.3    0.2   Var OE          NA           NA
```

Lastly, we perform assessment of the propensity score in IPW2 in the presence of interference.

```
# ------------------------------------------------------------
# 5. Assessing the fit of the generalized propensity scores
# ------------------------------------------------------------

### Check covariate balance for generalized propensity score: Fit a model on each covariate, then inclu


# IPW2 was fitted using "age10", so we check balance using "age10" (continuous covariate), not "agebin"
base_covariate=c("age10", "posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_age10","avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")
```

```r
##### check propensity score balance for IPW2
mat_1 = as.matrix(cbind(1, df[, row.names(coef(summary(M1)))[-1]])) # neighbors' treatment
mat_2 = as.matrix(cbind(1, df[, row.names(coef(summary(M2)))[-1]])) # Individual treatment

group_ps <- plogis(mat_1 %*% coef(M1))
ind_ps <- plogis(mat_2 %*% coef(M2))
df$group_ps_ipw2 <- group_ps
df$ind_ps_ipw2 <- ind_ps #Ke, should this be 1-ps for those not exposed?

### Step 1: check individual-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#       (1) ind_covariate ~ ind_treatment + ind_ps_ipw2;
#       (2) ind_covariate ~ ind_ps_ipw2
ipw2_balance <- data.frame()
for (i in 1:length(base_covariate)) {
  f1 <- paste0(base_covariate[i], " ~ ind_ps_ipw2")
  f2 <- paste0(base_covariate[i], " ~ treatment + ind_ps_ipw2")

  if (base_covariate[i] == "age10"){
    model_reduced <- glm(f1, data = df, family = gaussian)
    model_full    <- glm(f2, data = df, family = gaussian("identity"))
  } else {
    model_reduced <- glm(f1, data = df, family = binomial("logit"))
    model_full    <- glm(f2, data = df, family = binomial("logit"))
  }
  res_temp <- lmtest::lrtest(model_reduced, model_full)

  ipw2_balance <- rbind(ipw2_balance,
                        c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
  )
}


### Step 2: check group-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#       (1) group_covariate ~ group_treatment + group_ps;
#       (2) group_covariate ~ group_ps.

for (i in 1:length(avg_covariate)) {
  f1 <- paste0(avg_covariate[i], " ~ group_ps_ipw2")
  f2 <- paste0(avg_covariate[i], " ~ cbind(na_a, notna_a) + group_ps_ipw2")

  model_reduced <- glm(f1, data = df)
  model_full    <- glm(f2, data = df)

  res_temp <- lmtest::lrtest(model_reduced, model_full)

  ipw2_balance <- rbind(ipw2_balance,
                        c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
  )
}
colnames(ipw2_balance) <- c("Reduced model", "Full model", "Chisq", "Pr(>Chisq)")
ipw2_balance
```

```
##                     Reduced model
## 1              age10 ~ ind_ps_ipw2
## 2             posneg ~ ind_ps_ipw2
## 3               date ~ ind_ps_ipw2
## 4              share ~ ind_ps_ipw2
## 5          education ~ ind_ps_ipw2
## 6         employment ~ ind_ps_ipw2
## 7        avg_age10 ~ group_ps_ipw2
## 8       avg_posneg ~ group_ps_ipw2
## 9         avg_date ~ group_ps_ipw2
## 10       avg_share ~ group_ps_ipw2
## 11  avg_education ~ group_ps_ipw2
## 12 avg_employment ~ group_ps_ipw2
##                                                Full model  Chisq Pr(>Chisq)
## 1                        age10 ~ treatment + ind_ps_ipw2 0.0141     0.9053
## 2                       posneg ~ treatment + ind_ps_ipw2  5e-04     0.9815
## 3                         date ~ treatment + ind_ps_ipw2  3e-04     0.9873
## 4                        share ~ treatment + ind_ps_ipw2 0.0314     0.8593
## 5                    education ~ treatment + ind_ps_ipw2 0.0015     0.9687
## 6                   employment ~ treatment + ind_ps_ipw2 0.0055      0.941
## 7       avg_age10 ~ cbind(na_a, notna_a) + group_ps_ipw2 0.8063     0.6682
## 8       avg_posneg ~ cbind(na_a, notna_a) + group_ps_ipw2 0.5856     0.7462
## 9         avg_date ~ cbind(na_a, notna_a) + group_ps_ipw2 1.3213     0.5165
## 10       avg_share ~ cbind(na_a, notna_a) + group_ps_ipw2  5.956     0.0509
## 11  avg_education ~ cbind(na_a, notna_a) + group_ps_ipw2 0.3799      0.827
## 12 avg_employment ~ cbind(na_a, notna_a) + group_ps_ipw2 2.3496     0.3089
```

```r
#Optional: Format Latex Table for output of each estimation approach
#Create forest plot
library(dplyr)
library(forcats)
library(ggplot2)
library(knitr)
library(kableExtra)

z <- qnorm(0.975)

#####################################################
# 1. Result object helper
#####################################################
#library(conflicted)
#conflicts_prefer(dplyr::filter) # filter will now always use dplyr, even if MASS is loaded
#conflicts_prefer(dplyr::count)

make_result_object <- function(x) {
  if (is.list(x) && !is.data.frame(x)) {
    df_idx <- which(sapply(x, function(y) {
      is.data.frame(y) &&
        all(c("estimation", "alpha0", "alpha1", "type") %in% names(y))
    }))

    if (length(df_idx) == 0) {
      stop("No data frame with columns estimation, alpha0, alpha1, and type was found.")
    }
```

```
    x <- x[[df_idx[1]]]
  }

  if (!all(c("estimation", "alpha0", "alpha1", "type") %in% names(x))) {
    stop("Input must contain columns: estimation, alpha0, alpha1, type.")
  }

  est_rows <- x %>%
    filter(!grepl("^Var", type)) %>%
    mutate(
      effect_key = case_when(
        type == "Direct" ~ "DE",
        type == "Indirect" ~ "IE",
        type == "Total" ~ "TE",
        type == "Overall" ~ "OE",
        TRUE ~ type
      )
    )

  var_rows <- x %>%
    filter(grepl("^Var", type)) %>%
    mutate(
      effect_key = gsub("^Var\\s+", "", type),
      variance = estimation
    ) %>%
    select(alpha0, alpha1, effect_key, variance)

  results <- est_rows %>%
    left_join(
      var_rows,
      by = c("alpha0", "alpha1", "effect_key")
    ) %>%
    mutate(
      se = sqrt(variance),
      lower = estimation - z * se,
      upper = estimation + z * se
    ) %>%
    select(estimation, alpha0, alpha1, type, lower, upper)

  list(results = results)
}

##################################################
# 2. Standardize alpha display by estimator/effect
##################################################

prep_effect_results <- function(results, estimator_name) {
  out <- results %>%
    filter(!grepl("^Var", type)) %>%
    mutate(
      estimator = estimator_name,
      effect = case_when(
        type == "Direct" ~ "Direct",
```

```
        type == "Indirect" ~ "Indirect",
        type == "Total" ~ "Total",
        type == "Overall" ~ "Overall",
        TRUE ~ type
      )
    )

  # G-comp and DR store coverage columns reversed relative to display.
  if (estimator_name %in% c("G-comp", "DR")) {
    out <- out %>%
      mutate(
        raw_alpha1 = alpha0,
        raw_alpha0 = alpha1
      )
  } else {
    out <- out %>%
      mutate(
        raw_alpha1 = alpha1,
        raw_alpha0 = alpha0
      )
  }

  out %>%
    mutate(
      display_alpha1 = pmax(raw_alpha1, raw_alpha0),
      display_alpha0 = pmin(raw_alpha1, raw_alpha0),
      contrast = paste0("(", display_alpha1, ", ", display_alpha0, ")")
    ) %>%
    filter(
      case_when(
        effect == "Direct" ~ display_alpha1 == display_alpha0,
        effect %in% c("Indirect", "Total") ~ display_alpha1 > display_alpha0,
        effect == "Overall" ~ TRUE,
        TRUE ~ TRUE
      )
    )
}

##################################################
# 3. LaTeX table helper
##################################################

make_latex_effect_table <- function(results,
                                    estimator_name,
                                    caption = "Estimated causal effects",
                                    digits = 3) {
  table_df <- prep_effect_results(results, estimator_name) %>%
    mutate(
      effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall")),
      estimate = round(estimation, digits),
      ci = ifelse(
        is.na(lower) | is.na(upper),
        "",
```

```
        paste0(
          "(", round(lower, digits), ", ",
          round(upper, digits), ")"
        )
      )
    ) %>%
    arrange(effect, desc(display_alpha1), desc(display_alpha0)) %>%
    select(
      Effect = effect,
      `$(\\alpha_1, \\alpha_0)$` = contrast,
      `Point estimate` = estimate,
      `95\\% CI` = ci
    )

  direct_n <- sum(table_df$Effect == "Direct")
  indirect_n <- sum(table_df$Effect == "Indirect")
  total_n <- sum(table_df$Effect == "Total")
  overall_n <- sum(table_df$Effect == "Overall")

  tab <- table_df %>%
    kable(
      format = "latex",
      booktabs = TRUE,
      escape = FALSE,
      caption = caption,
      align = c("l", "c", "c", "c")
    ) %>%
    kable_styling(latex_options = c("hold_position"))

  start <- 1

  if (direct_n > 0) {
    tab <- tab %>% pack_rows("Direct effects", start, start + direct_n - 1, italic = TRUE)
    start <- start + direct_n
  }

  if (indirect_n > 0) {
    tab <- tab %>% pack_rows("Indirect effects", start, start + indirect_n - 1, italic = TRUE)
    start <- start + indirect_n
  }

  if (total_n > 0) {
    tab <- tab %>% pack_rows("Total effects", start, start + total_n - 1, italic = TRUE)
    start <- start + total_n
  }

  if (overall_n > 0) {
    tab <- tab %>% pack_rows("Overall effects", start, start + overall_n - 1, italic = TRUE)
  }

  tab
}
```

```r
################################################
# 4. Create result objects
################################################

res_ipw1 <- make_result_object(IPW1)
res_ipw2 <- make_result_object(IPW2)

res_gcomp <- if (exists("res_gcomp") && "results" %in% names(res_gcomp)) {
  res_gcomp
} else {
  make_result_object(gcomp_table)
}

res_dr <- if (exists("res_dr") && "results" %in% names(res_dr)) {
  res_dr
} else {
  make_result_object(dr_table)
}

################################################
# 5. Create LaTeX tables
################################################

ipw1_tab <- make_latex_effect_table(
  res_ipw1$results,
  estimator_name = "IPW1",
  caption = "IPW1 estimates of causal effects"
)

ipw2_tab <- make_latex_effect_table(
  res_ipw2$results,
  estimator_name = "IPW2",
  caption = "IPW2 estimates of causal effects"
)

gcomp_tab <- make_latex_effect_table(
  res_gcomp$results,
  estimator_name = "G-comp",
  caption = "G-computation estimates of causal effects"
)

dr_tab <- make_latex_effect_table(
  res_dr$results,
  estimator_name = "DR",
  caption = "Doubly robust estimates of causal effects"
)

cat(ipw1_tab, file = "ipw1_effects_table.tex")
cat(ipw2_tab, file = "ipw2_effects_table.tex")
cat(gcomp_tab, file = "gcomp_effects_table.tex")
cat(dr_tab, file = "dr_effects_table.tex")

################################################
```

```r
# 6. Forest plot data
#####################################################

estimator_levels <- c("IPW1", "IPW2", "G-comp", "DR")

df <- bind_rows(
  prep_effect_results(res_ipw1$results, "IPW1"),
  prep_effect_results(res_ipw2$results, "IPW2"),
  prep_effect_results(res_gcomp$results, "G-comp"),
  prep_effect_results(res_dr$results, "DR")
) %>%
  mutate(
    estimator = factor(estimator, levels = estimator_levels),
    effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall"))
  )

# Optional check: verify all estimator/effect/contrast combinations are present.
df %>%
  dplyr::count(estimator, effect, contrast) %>%
  arrange(estimator, effect, contrast)
```

```
##    estimator   effect   contrast n
## 1       IPW1   Direct (0.2, 0.2) 1
## 2       IPW1   Direct (0.3, 0.3) 1
## 3       IPW1   Direct (0.4, 0.4) 1
## 4       IPW1   Direct (0.5, 0.5) 1
## 5       IPW1 Indirect (0.3, 0.2) 1
## 6       IPW1 Indirect (0.4, 0.2) 1
## 7       IPW1 Indirect (0.4, 0.3) 1
## 8       IPW1 Indirect (0.5, 0.2) 1
## 9       IPW1 Indirect (0.5, 0.3) 1
## 10      IPW1 Indirect (0.5, 0.4) 1
## 11      IPW1    Total (0.3, 0.2) 1
## 12      IPW1    Total (0.4, 0.2) 1
## 13      IPW1    Total (0.4, 0.3) 1
## 14      IPW1    Total (0.5, 0.2) 1
## 15      IPW1    Total (0.5, 0.3) 1
## 16      IPW1    Total (0.5, 0.4) 1
## 17      IPW1  Overall (0.3, 0.2) 1
## 18      IPW1  Overall (0.4, 0.2) 1
## 19      IPW1  Overall (0.4, 0.3) 1
## 20      IPW1  Overall (0.5, 0.2) 1
## 21      IPW1  Overall (0.5, 0.3) 1
## 22      IPW1  Overall (0.5, 0.4) 1
## 23      IPW2   Direct (0.2, 0.2) 1
## 24      IPW2   Direct (0.3, 0.3) 1
## 25      IPW2   Direct (0.4, 0.4) 1
## 26      IPW2   Direct (0.5, 0.5) 1
## 27      IPW2 Indirect (0.3, 0.2) 1
## 28      IPW2 Indirect (0.4, 0.2) 1
## 29      IPW2 Indirect (0.4, 0.3) 1
## 30      IPW2 Indirect (0.5, 0.2) 1
## 31      IPW2 Indirect (0.5, 0.3) 1
```

```
## 32      IPW2 Indirect (0.5, 0.4) 1
## 33      IPW2    Total (0.3, 0.2) 1
## 34      IPW2    Total (0.4, 0.2) 1
## 35      IPW2    Total (0.4, 0.3) 1
## 36      IPW2    Total (0.5, 0.2) 1
## 37      IPW2    Total (0.5, 0.3) 1
## 38      IPW2    Total (0.5, 0.4) 1
## 39      IPW2  Overall (0.3, 0.2) 1
## 40      IPW2  Overall (0.4, 0.2) 1
## 41      IPW2  Overall (0.4, 0.3) 1
## 42      IPW2  Overall (0.5, 0.2) 1
## 43      IPW2  Overall (0.5, 0.3) 1
## 44      IPW2  Overall (0.5, 0.4) 1
## 45    G-comp   Direct (0.2, 0.2) 1
## 46    G-comp   Direct (0.3, 0.3) 1
## 47    G-comp   Direct (0.4, 0.4) 1
## 48    G-comp   Direct (0.5, 0.5) 1
## 49    G-comp Indirect (0.3, 0.2) 1
## 50    G-comp Indirect (0.4, 0.2) 1
## 51    G-comp Indirect (0.4, 0.3) 1
## 52    G-comp Indirect (0.5, 0.2) 1
## 53    G-comp Indirect (0.5, 0.3) 1
## 54    G-comp Indirect (0.5, 0.4) 1
## 55    G-comp    Total (0.3, 0.2) 1
## 56    G-comp    Total (0.4, 0.2) 1
## 57    G-comp    Total (0.4, 0.3) 1
## 58    G-comp    Total (0.5, 0.2) 1
## 59    G-comp    Total (0.5, 0.3) 1
## 60    G-comp    Total (0.5, 0.4) 1
## 61    G-comp  Overall (0.3, 0.2) 1
## 62    G-comp  Overall (0.4, 0.2) 1
## 63    G-comp  Overall (0.4, 0.3) 1
## 64    G-comp  Overall (0.5, 0.2) 1
## 65    G-comp  Overall (0.5, 0.3) 1
## 66    G-comp  Overall (0.5, 0.4) 1
## 67        DR   Direct (0.2, 0.2) 1
## 68        DR   Direct (0.3, 0.3) 1
## 69        DR   Direct (0.4, 0.4) 1
## 70        DR   Direct (0.5, 0.5) 1
## 71        DR Indirect (0.3, 0.2) 1
## 72        DR Indirect (0.4, 0.2) 1
## 73        DR Indirect (0.4, 0.3) 1
## 74        DR Indirect (0.5, 0.2) 1
## 75        DR Indirect (0.5, 0.3) 1
## 76        DR Indirect (0.5, 0.4) 1
## 77        DR    Total (0.3, 0.2) 1
## 78        DR    Total (0.4, 0.2) 1
## 79        DR    Total (0.4, 0.3) 1
## 80        DR    Total (0.5, 0.2) 1
## 81        DR    Total (0.5, 0.3) 1
## 82        DR    Total (0.5, 0.4) 1
## 83        DR  Overall (0.3, 0.2) 1
## 84        DR  Overall (0.4, 0.2) 1
## 85        DR  Overall (0.4, 0.3) 1
```

```
## 86        DR  Overall (0.5, 0.2) 1
## 87        DR  Overall (0.5, 0.3) 1
## 88        DR  Overall (0.5, 0.4) 1
```

```
################################################
# 7. Forest plot helper
################################################

estimator_colors <- c(
  "IPW1" = "#009E73",
  "IPW2" = "#0072B2",
  "G-comp" = "#56B4E9",
  "DR" = "#E69F00"
)

estimator_shapes <- c(
  "IPW1" = 15,
  "IPW2" = 16,
  "G-comp" = 17,
  "DR" = 18
)

estimator_offsets <- c(
  "IPW1" = 0.27,
  "IPW2" = 0.09,
  "G-comp" = -0.09,
  "DR" = -0.27
)

### specify x-axis range in the plot
x_min <- min(df$lower) - 0.5
x_max <- max(df$upper) + 0.1

make_effect_forest_plot <- function(plot_data, effect_name) {
  plot_data <- plot_data %>%
    filter(effect == effect_name)

  contrast_map <- plot_data %>%
    distinct(contrast, display_alpha1, display_alpha0) %>%
    arrange(desc(display_alpha1), desc(display_alpha0)) %>%
    dplyr::mutate(contrast_id = row_number())

  plot_data <- plot_data %>%
    left_join(
      contrast_map,
      by = c("contrast", "display_alpha1", "display_alpha0")
    ) %>%
    dplyr::mutate(
      y_pos = contrast_id + estimator_offsets[as.character(estimator)]
    )

  ggplot(
    plot_data,
    aes(color = estimator, shape = estimator)
```

```
  ) +
    geom_vline(
      xintercept = 0,
      linetype = "dashed",
      linewidth = 0.6,
      color = "gray30"
    ) +
    geom_segment(
      aes(x = lower, xend = upper, y = y_pos, yend = y_pos),
      linewidth = 0.8
    ) +
    geom_point(
      aes(x = estimation, y = y_pos),
      size = 2.2
    ) +
    scale_x_continuous(limits=c(x_min, x_max)) +
    scale_y_continuous(
      breaks = contrast_map$contrast_id,
      labels = contrast_map$contrast
    ) +
    scale_color_manual(values = estimator_colors, breaks = estimator_levels) +
    scale_shape_manual(values = estimator_shapes, breaks = estimator_levels) +
    labs(
      title = paste(effect_name, "Effects"),
      x = "Estimate (95% CI)",
      y = expression(paste("Allocation (", alpha[1], ", ", alpha[0], ")")),
      color = "Estimator",
      shape = "Estimator"
    ) +
    theme_minimal(base_size = 16) +
    theme(
      legend.position = "top",
      plot.title = element_text(face = "bold", hjust = 0),
      axis.text.y = element_text(size = 12),
      panel.grid.minor = element_blank(),
      panel.grid.major.y = element_blank()
    )
}

################################################
# 8. Create four forest plots
################################################

p_direct <- make_effect_forest_plot(df, "Direct")
p_indirect <- make_effect_forest_plot(df, "Indirect")
p_total <- make_effect_forest_plot(df, "Total")
p_overall <- make_effect_forest_plot(df, "Overall")

ggsave("spillover_forest_direct.pdf", p_direct, width = 10, height = 4.5)
ggsave("spillover_forest_indirect.pdf", p_indirect, width = 10, height = 5.5)
ggsave("spillover_forest_total.pdf", p_total, width = 10, height = 5.5)
ggsave("spillover_forest_overall.pdf", p_overall, width = 10, height = 5.5)
```